

Threat Intelligence Lookup & IOC Verification — Investigation Report

1. Introduction

Threat intelligence is an important part of cybersecurity because it helps security analysts identify, investigate, and classify potentially malicious indicators. Indicators of Compromise (IOCs) such as IP addresses, domains, and file hashes can provide evidence of suspicious or malicious activity.

This project focuses on the investigation and verification of selected IOCs using publicly available threat intelligence platforms. The investigated indicators were checked using ThreatFox, VirusTotal, URLScan.io, and MITRE ATT&CK. The investigation results were documented and classified based on the available evidence.

2. Project Objective

The main objectives of this project are:

- To collect and investigate suspicious IP addresses, domains, and file hashes.
 - To verify IOC reputation using public threat intelligence platforms.
 - To identify malware or threat families associated with suspicious indicators.
 - To map relevant threat activity to MITRE ATT&CK techniques where applicable.
 - To document investigation results in a structured IOC log.
 - To distinguish malicious indicators from benign security-testing indicators.
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3. Tools and Technologies Used

ThreatFox

ThreatFox was used to identify threat intelligence information associated with IP addresses and domains, including malware families, threat types, confidence levels, and first-seen information.

VirusTotal

VirusTotal was used to examine the reputation of IP addresses, domains, and file hashes using multiple security vendors and analysis engines.

URLScan.io

URLScan.io was used to investigate URLs and obtain information about website scanning, network activity, and scan results.

MITRE ATT&CK

MITRE ATT&CK was used to identify relevant attacker techniques associated with the investigated malware and threat behavior.

Microsoft Excel

Microsoft Excel was used to maintain the structured IOC investigation log.

4. Investigation Methodology

The investigation followed these steps:

1. Select suspicious indicators from public threat intelligence sources and sample data.
2. Identify the IOC type.
3. Search the IOC in ThreatFox where applicable.
4. Verify the IOC using VirusTotal.
5. Perform URLScan.io analysis where applicable.
6. Identify the associated malware or threat family.
7. Map relevant activity to MITRE ATT&CK techniques.
8. Record the results in the IOC log.
9. Classify each IOC based on the collected evidence.

The investigation was performed using lookup and analysis techniques. Suspicious URLs and files were not downloaded or executed.

5. IOC Investigation — IP Address

IOC

124.223.177.82:22

IOC Type

IP Address / Port

ThreatFox Findings

The IP address and port were identified in ThreatFox with the following information:

- IOC Type: ip:port
- Threat Type: botnet_cc
- Malware: **Cobalt Strike**
- Confidence Level: **100%**
- Is Compromised: **True**
- ASN: **AS45090**
- Country: **China (CN)**
- Tag: **CobaltStrike**
- First Seen: **2026-09-21 14:05:08 UTC**

The ThreatFox information associates the indicator with Cobalt Strike activity.

VirusTotal Findings

The IP address was also investigated using VirusTotal.

- Detection: **5/89**
- ASN: **45090**
- ASN Organization: **Shenzhen Tencent Computer Systems Company Limited**
- Network: 124.220.0.0/14
- Network Name: **TencentCloud**
- Country: **CN**
- RIR: **APNIC**

The VirusTotal result showed detections from multiple security engines, providing additional evidence that the indicator requires investigation.

URLScan.io Findings

The address was checked using URLScan.io.

The scan could not successfully retrieve the website and returned:

net::ERR_EMPTY_RESPONSE

This means that the scanning service did not receive a usable response from the target.

Classification

Malicious

The classification is based on the ThreatFox association with Cobalt Strike, 100% confidence, compromised status, and supporting VirusTotal detections.

6. IOC Investigation — Domain

IOC

h67as5d5x.m6p3wca1.cc

IOC Type

Domain

ThreatFox Findings

The domain was associated with **Cobalt Strike** activity in ThreatFox.

The recorded confidence level was:

75%

This indicates that the domain had a documented association with the identified threat activity.

VirusTotal Findings

VirusTotal analysis showed:

- Detection: **15/91**
- Category: Suspicious
- Multiple security vendors classified the domain as related to phishing or fraud.
- Final URL: <http://h67as5d5x.m6p3wca1.cc/>
- HTTP Status: **200**
- Content Type: **image/png**
- HTML Title: **Not Found**

The VirusTotal results showed multiple independent security-engine detections.

URLScan.io Findings

URLScan.io was used to investigate the domain.

The scan was unsuccessful because the website could not be contacted successfully. The reported reasons included possible DNS or network issues and other connection-related problems.

Classification

Malicious

The classification is based on the ThreatFox association with Cobalt Strike and the multiple VirusTotal detections and suspicious classifications.

7. IOC Investigation — SHA-256 Hash

IOC

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

IOC Type

SHA-256 File Hash

VirusTotal Findings

The hash was investigated using VirusTotal.

The results included:

- Detection: **66/68**
- File Size: **68 bytes**
- File Type: **PowerShell**
- Magic: **EICAR virus test files**
- TrID Identification: **EICAR antivirus test file**

- First Submission: **2006-05-22**
- First Seen in the Wild: **2012-06-14**

The associated MD5 hash was:

44d88612fea8a8f36de82e1278abb02f

The associated SHA-1 hash was:

3395856ce81f2b7382dee72602f798b642f14140

Important Finding

Although VirusTotal reported a high detection rate, the file is the **EICAR antivirus test file**.

EICAR is intentionally designed to test antivirus and security products. It is not intended to represent an actual malicious infection.

Classification

Security Test / Benign

The indicator was therefore classified separately from the two malicious IOCs.

8. MITRE ATT&CK Mapping

The Cobalt Strike-related indicators were mapped to the following MITRE ATT&CK technique:

T1021.004 — Remote Services: SSH

Cobalt Strike has documented capabilities associated with remote services, including SSH.

This technique was therefore used as the relevant MITRE ATT&CK mapping for the Cobalt Strike-associated indicators investigated in this project.

Mapping

IOC	Malware/Family	MITRE ID	Technique
124.223.177.82:22	Cobalt Strike	T1021.004	Remote Services: SSH
h67as5d5x.m6p3wca1.cc	Cobalt Strike	T1021.004	Remote Services: SSH
EICAR SHA-256	EICAR Test File	N/A	N/A

The EICAR test file was not mapped to an attacker technique because it is a security testing file rather than an actual malware sample.

9. IOC Investigation Summary

IOC	Type	Source	Malware /Family	Virus Total	URLScan	MITRE ID	Classification
124.223.177.82:22	IP:Port	ThreatFox	Cobalt Strike	5/89	ERR_EMPTY_RESPONSE	T102.1.004	Malicious
h67as5d5x.m6p3wca1.cc	Domain	ThreatFox	Cobalt Strike	15/91	Could not scan	T102.1.004	Malicious
275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f	SHA-256	VirusTotal	EICAR Test File	66/68	N/A	N/A	Security Test / Benign

10. Investigation Findings

The investigation produced the following findings:

1. The IP address 124.223.177.82:22 was associated with Cobalt Strike in ThreatFox and had a 100% confidence rating in the recorded result.
2. The domain h67as5d5x.m6p3wca1.cc was associated with Cobalt Strike and received multiple detections from VirusTotal security vendors.
3. Both Cobalt Strike-associated indicators were mapped to **T1021.004 — Remote Services: SSH** as the relevant MITRE ATT&CK technique.
4. The SHA-256 hash 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f produced a high VirusTotal detection rate, but the file was identified as the **EICAR antivirus test file**.
5. The investigation demonstrates why IOC classification should not rely only on a detection count. Context from threat intelligence sources and file identification is also important.

11. Security Recommendations

Based on the investigation workflow, the following practices are recommended:

- Verify suspicious IOCs using multiple threat intelligence sources.
- Maintain an organized IOC database or spreadsheet.
- Record the source and date of IOC observations.
- Investigate suspicious IP addresses and domains before allowing network communication.
- Monitor systems for connections to known malicious infrastructure.
- Use MITRE ATT&CK to understand the techniques associated with identified threats.

- Avoid downloading or executing suspicious files during IOC investigation.
 - Use controlled security-testing files such as EICAR only in appropriate testing environments.
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12. Limitations

Threat intelligence results can change over time. Detection counts may increase or decrease as security vendors update their databases.

A failed URLScan result does not by itself prove that an indicator is malicious or benign. Similarly, an IP address being hosted by a legitimate cloud provider does not automatically determine the reputation of the specific IP.

Therefore, IOC classification in this project was based on the combination of available threat intelligence evidence rather than a single source.

13. Conclusion

This project demonstrated a practical workflow for investigating and verifying Indicators of Compromise using publicly available cybersecurity tools.

Three different IOC types were investigated: an IP/port, a domain, and a SHA-256 file hash. ThreatFox, VirusTotal, URLScan.io, and MITRE ATT&CK were used to collect and correlate information.

The first two indicators were associated with Cobalt Strike and classified as malicious based on the collected threat intelligence evidence. The third indicator was identified as the EICAR antivirus test file and was classified as a benign security-testing indicator.

The final results were documented in an IOC log and supported with screenshots of the investigation process. This project demonstrates basic threat intelligence, IOC verification, reputation analysis, and MITRE ATT&CK mapping skills.

14. Project Files

The GitHub repository contains:

Threat-Intelligence-IOC-Verification/

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|— README.md

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|— screenshots/

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|— 01-threat-intelligence.png

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|— 02-virus-total-analysis.png

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|— 03-urlscan-analysis.png

- | | └─ 04-mitre-attack.png
- | | └─ 05-ioc-results.png
- |
- | └─ ioc-log/
- | | └─ Threat_Intelligence_IOC_Log.xlsx
- |
- | └─ report/
- | | └─ Threat_Intelligence_IOC_Investigation_Report.pdf

15. Safety Note

This investigation was performed for cybersecurity learning and defensive threat intelligence purposes. The investigated indicators were analyzed through reputation and intelligence lookup services. Suspicious files and URLs were not intentionally downloaded or executed.

End of Report